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Image display device

Abstract

An image display device includes a display element, an optical member for imaging, a circuit board, an optical case configured to accommodate the display element and the optical member, a metallic frame configured to support the circuit board from below, and an exterior case configured to accommodate the optical case and the metallic frame, wherein the metallic frame is disposed above the optical case, and the exterior case includes an upper case and a lower case, and includes a first seal member at a first facing portion between the upper case and the lower case.

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2022-130279, filed Aug. 17, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to an image display device that enables observation of a virtual image.

2. Related Art

(3) JP-A-2017-147523 discloses an image display device in which an image projection unit and a control unit are accommodated in a case member mounted in a frame member, and the control unit is disposed below the image projection unit. In this device, the control unit is surrounded by a first rib of an inner case located between the image projection unit and the control unit, and is further covered by an exterior case. In this device, the image projection unit and the control unit are waterproofed by the exterior case, and the control unit is waterproofed by the inner case.

(4) In the device of the background art described above, waterproofing is achieved by fitting the inner case and the exterior case together, but, when fastening of the exterior case or the like is loosened, there is a possibility that a waterproof function will deteriorate in a fitting portion.

SUMMARY

(5) According to an aspect of the present disclosure, there is provided an image display device including a display element, an optical member for imaging, a circuit board, an optical case configured to accommodate the display element and the optical member, a metallic frame configured to support the circuit board from below, and an exterior case configured to accommodate the optical case and the metallic frame, wherein the metallic frame is disposed above the optical case, and the exterior case includes an upper case and a lower case, and includes a first seal member at a first facing portion between the upper case and the lower case.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an external perspective view for describing a mounting state of an image display device according to an embodiment.
- (2) FIG. 2 is a front view of the image display device.
- (3) FIG. 3 is a perspective view of the image display device from diagonally behind.
- (4) FIG. 4 is a side cross-sectional view for describing an optical internal structure of the display device on one side.
- (5) FIG. 5 is an exploded perspective view of the image display device.
- (6) FIG. 6 is a perspective view illustrating an optical device.
- (7) FIG. 7 is a perspective view illustrating a state in which a shield member is removed from the optical device of FIG. 6.
- (8) FIG. 8 is a perspective view of a pair of optical units.
- (9) FIG. 9 is a plan view and a perspective view for describing a metallic frame supporting a display unit.
- (10) FIG. 10 is a perspective view for describing a modified example of a support structure and the like.
- (11) FIG. 11 is a plan view of an upper cover, a plan view of a middle cover, and a plan view of a lower cover.
- (12) FIG. 12 is a side view for describing a change in posture of the optical device.
- (13) FIG. 13 is a perspective view of the back side of the image display device.
- (14) FIG. 14 is a side cross-sectional view of a barrel, an optical member held by the barrel, and the like.
- (15) FIG. 15 is an exploded perspective view of the barrel.
- (16) FIG. 16 is a perspective view and a side cross-sectional view of the front side, and a perspective view of the back side illustrating a display unit.
- (17) FIG. 17 is an enlarged cross-sectional view of a barrel front portion.
- (18) FIG. 18 is a plan view of an optical unit.
- (19) FIG. 19 is a front view and a plan view of an optical unit.
- (20) FIG. 20 is a side cross-sectional perspective view for describing an internal structure of the image display device.
- (21) FIG. 21 is a side cross-sectional perspective view for describing the internal structure of the image display device.
- (22) FIG. 22 is a side cross-sectional perspective view illustrating a cross-sectional structure of an image display device according to a modified example.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

- (23) An embodiment of an image display device according to the present disclosure will be described below with reference to FIGS. 1, 2 and the like.
- (24) FIG. 1 is a perspective view illustrating a wearing state of a head-mounted display (hereinafter, also referred to as an HMD) 200, FIG. 2 is a front view of a main body of the HMD

200, and FIG. **3** is a perspective view of the main body of the HMD **200** seen from the rear and above. The HMD **200** allows an observer or a wearer US who wears the HMD **200** to recognize a video as a virtual image. In FIGS. **1**, **2** and the like, X, Y, and Z are of an orthogonal coordinate system, a +X direction corresponds to a horizontal direction in which both eyes EY of the observer or wearer US wearing the HMD **200** or an image display device **100** are arranged, a +Y direction corresponds to an upward direction orthogonal to the horizontal direction in which both eyes EY of the wearer US are arranged, and a +Z direction corresponds to a forward or frontal direction with respect to the wearer US. The $\pm Y$ directions are parallel to a vertical axis or a vertical direction.

(25) The HMD **200** includes a first display device **100A** for the right eye, a second display device **100B** for the left eye, a pair of temple type support devices **100C** that support the display devices **100A** and **100B**, and a user terminal **90** as an information terminal. The first display device **100A** functions independently as an image display device, and is constituted of a first display driving unit **102a** disposed at an upper portion thereof, a first combiner **103a** that is shaped like a spectacle lens and covers the front of the eye, and a light transmission cover **104a** that covers the combiner **103a** from the front. Similarly, the second display device **100B** functions independently as an image display device, and is constituted of a second display driving unit **102b** disposed at an upper portion thereof, a second combiner **103b** that is shaped like a spectacle lens and covers the front of the eye, and a light transmission cover **104b** that covers the combiner **103b** from the front. The support device **100C** is a mounting member that is mounted on the head of the wearer US, and supports the upper end sides of the pair of combiners **103a** and **103b** and the upper end sides of the pair of light transmission covers **104a** and **104b** via the display driving units **102a** and **102b** integrated in appearance. In this specification, one of the first combiner **103a** and the second combiner **103b** may be referred to as the combiner **103**. A combination of the pair of display driving units **102a** and **102b** is referred to as a driving device **102**. A combination of the pair of light transmission covers **104a** and **104b** is referred to as a shade **104**.

(26) The first display device **100A** and the second display device **100B** are optically identical or mirror-reversed, and a detailed description of the second display device **100B** will be omitted.

(27) Referring to FIGS. **2** and **3**, the driving device **102** includes a front cover **71**, a middle frame **72**, and a lower cover **73** as an exterior case **7** or a cover member **70** constituting an exterior. The front cover **71** is an upper case CV1 that covers a front portion and an upper portion of the driving device **102**. The front cover **71** and the shade **104** are joined and integrated. The middle frame **72** is one of members constituting the lower case CV2 that supports optical members and the like incorporated in the driving device **102** at both ends and covers the incorporated components mainly from the front side. The lower cover **73** is one of members constituting the lower case CV2 that covers the optical members incorporated in the driving device **102** mainly from the rear and below. Hinges **74** are mounted on both ends of the middle frame **72**, and support the pair of support devices **100C** in a foldable manner.

(28) FIG. **4** is a side cross-sectional view illustrating an optical structure of the first display device **100A**. The first display device **100A** includes a first display element **11a**, a first display unit **20a**, and a first circuit member **80a**. The first display element **11a** is an image light generating device. The first display unit **20a** is an imaging optical system that forms a virtual image, and includes a projection lens **21**, a prism mirror **22**, and a see-through mirror **23** in an integrated state. In the first display unit **20a**, the projection lens **21** and the prism mirror **22** function as a first projection optical system **12a** on which image light ML from the first display element **11a** is incident, and the see-through mirror **23** functions as a partially transmissive mirror **123** that partially reflects the image light ML emitted from the first projection optical system **12a** toward a pupil position PP or an eye EY. The projection lens **21** and the prism mirror **22** constituting the first projection optical system **12a** correspond to a first optical member and a second optical member on which the video light or the image light ML is incident, respectively. Further, the first display element **11a**, the projection lens **21**, and the prism mirror **22** are optical elements that constitute the first display driving unit

102a illustrated in FIG. 1, and the see-through mirror **23** corresponds to the first combiner **103a** illustrated in FIG. 1. The see-through mirror **23** has an exterior that is convex outward. The projection lens **21** and the prism mirror **22** constituting the first projection optical system **12a** are fixed in the barrel **41** in a mutually aligned state together with the first display element **11a**. The barrel **41** is an optical case CA in which the first projection optical system **12a** is accommodated in a positioned state.

(29) The barrel **41** or the optical case CA that accommodates the optical elements constituting the projection lens **21** and the like is supported by a first metallic frame **52a** and is disposed below the first metallic frame **52a**. The first metallic frame **52a** is covered by the cover member **70**, and the barrel **41** is also entirely covered by the cover member **70**. The first metallic frame **52a** is formed of a metallic material. The barrel **41** and the cover member **70** are formed of a light-shielding resin material, and one surface of the prism mirror **22** is exposed at an emission port **41o** of the barrel **41**. The barrel **41** is in contact with the first metallic frame **52a** so that an upper portion **41t** is fitted thereto, and is fixed in a state in which it is suspended from the first metallic frame **52a**. As a result, the first display unit **20a** is fixed in a state in which it is suspended from the first metallic frame **52a** by the upper portion **41t** of the barrel **41** being brought into contact with and screwed to the first metallic frame **52a** so as to be fitted thereto. The first metallic frame **52a** has a recess RE for arranging the first circuit member **80a** on an upper side thereof. The cover member **70** has an internal space ES of which a sealing property is enhanced by being combined with the barrel **41** or the like, and an accessory component circuit member **80c** is accommodated above the first circuit member **80a** in the internal space ES.

(30) In the first display device **100A**, the first display element **11a** is a spontaneous light emission type image light generation device. The first display element **11a** emits the image light ML to the first projection optical system **12a**. The barrel **41** accommodates and supports the first display element **11a** together with optical elements such as the projection lens **21**. The first display element **11a** is, for example, an organic electroluminescence (EL) display, and forms a color still image or moving image on a two-dimensional display surface **11d**. The first display element **11a** is driven by the first circuit member **80a**, specifically a display control device **88**, to perform a display operation. The first display element **11a** is not limited to the organic EL display, and can be replaced with a display device using inorganic EL, an organic LED, an LED array, a laser array, a quantum dot light emission element, or the like. The first display element **11a** is not limited to the spontaneous light emission type image light generation device, and may include an LCD and another light modulation element, and may form an image by illuminating the light modulation element with a light source such as a backlight. As the first display element **11a**, a liquid crystal on silicon (LCOS) (LCoS is a registered trademark), a digital micro-mirror device, or the like may be used instead of an LCD.

(31) In the present embodiment, the first display unit **20a** includes two reflection surfaces, and an optical path is bent by the see-through mirror **23** and the prism mirror **22**. The first display unit **20a** is an off-axis optical system OS. The projection lens **21**, the prism mirror **22**, and the see-through mirror **23** are disposed to be non-axially symmetrical and have an optical surface that is non-axisymmetric. In the first display unit **20a**, the optical elements **21**, **22**, and **23** are arranged along an off-axis plane by bending an optical axis AX within an off-axis plane (that is, a reference plane) parallel to an YZ plane. In other words, in the off-axis plane (the reference plane) parallel to the YZ plane, an optical path portion P1 from the projection lens **21** to an inner reflection surface **22b**, an optical path portion P2 from the inner reflection surface **22b** to the see-through mirror **23**, and an optical path portion P3 from the see-through mirror **23** to the pupil position PP are bent in a Z shape in two stages. As a result, a normal line at a central portion in which the optical axis AX in the see-through mirror **23** intersects forms an angle of about $\theta=40$ to 50° with respect to the Z direction. In the first display unit **20a**, the optical elements **21**, **22**, and **23** constituting the first display device **100A** are arranged so that height positions thereof change in a longitudinal direction,

and an increase in a transverse width of the first display device **100A** can be prevented. Further, since the optical path portions **P1** to **P3** are disposed to be folded in two stages in a Z shape by folding of the optical path due to reflection by the prism mirror **22** or the like, and since the optical path portions **P1** and **P3** are relatively close to horizontal, it is possible to reduce a size of the first display unit **20a** also in the vertical direction and the forward and rearward direction. In addition, since an inclination angle θ of the central portion of the see-through mirror **23** is 40° to 50° , when an inclination of the optical path portion **P3** corresponding to the line of sight is constant, the inclination of the optical path portion **P2** with respect to the Z-axis is 70° to 90° , and a thickness of the image display device **100** in the Z direction can be easily reduced.

(32) In the first display unit **20a**, the optical path portion **P1** from the projection lens **21** to the inner reflection surface **22b** extends rearward in a slightly obliquely upward direction or a direction substantially parallel to the Z direction with respect to the viewpoint. The optical path portion **P2** from the inner reflection surface **22b** to the see-through mirror **23** extends forward in an obliquely downward direction. When a horizontal plane direction (an XZ plane) is used as a reference, the inclination of the optical path portion **P2** is larger than the inclination of the optical path portion **P1**. The optical path portion **P3** from the see-through mirror **23** to the pupil position PP extends rearward in a slightly obliquely upward direction or in a direction substantially parallel to the Z direction. In the illustrated example, a portion of the optical axis AX corresponding to the optical path portion **P3** is about -10° in the +Z direction with a downward direction being negative. That is, the partially transmissive mirror **123** reflects the image light ML so that the optical axis AX or the optical path portion **P3** is directed upward by a predetermined angle, that is, upward by about 10° . As a result, an emission optical axis EX which is an extension of a portion of the optical axis AX corresponding to the optical path portion **P3** extends obliquely downward by about 10° with respect to a central axis HX parallel to the forward +Z direction. This is because the line of sight of a human being is stable in a slightly lowered eye state in which the line of sight is inclined downward by approximately 10° with respect to a horizontal direction. The central axis HX that extends in the horizontal direction with respect to the pupil position PP assumes a case in which the wearer US wearing the first display device **100A** relaxes in an upright posture and faces the front and gazes in the horizontal direction or in a horizontal line.

(33) In the first display unit **20a**, the projection lens **21** includes a first lens **21o**, a second lens **21p**, and a third lens **21q**. The projection lens **21** receives the image light ML emitted from the first display element **11a** and causes it to be incident on the prism mirror **22**. The projection lens **21** focuses the image light ML emitted from the first display element **11a** into a state close to a parallel luminous flux. Optical surfaces of the first lens **21o**, the second lens **21p**, and the third lens **21q** constituting the projection lens **21**, that is, an incident surfaces and the emission surface are free curved surfaces or aspherical surfaces, have asymmetry with respect to the longitudinal direction parallel to the YZ plane and intersecting the optical axis AX with the optical axis AX interposed therebetween, and have symmetry with respect to the transverse direction or the X direction with the optical axis AX interposed therebetween. The first lens **21o**, the second lens **21p**, and the third lens **21q** are formed of a resin, for example, but can also be formed of glass. An antireflection film can be formed at the optical surfaces of the first lens **21o**, the second lens **21p**, and the third lens **21q** constituting the projection lens **21**.

(34) The prism mirror **22** is an optical member having a refracting and reflecting function that combines the functions of a mirror and a lens, and refracts and reflects the image light ML from the projection lens **21**. The prism mirror **22** has an incident surface **22a** corresponding to an incident portion, an inner reflection surface **22b** as a reflection surface corresponding to an reflection portion, and an emission surface **22c** corresponding to an emission portion. The prism mirror **22** emits the image light ML incident from the front so that it is bent in a direction inclined downward with respect to a direction in which an incident direction is reversed (a direction of the light source seen from the prism mirror **22**). The incident surface **22a**, the inner reflection surface **22b**, and the

emission surface **22c** which are the optical surfaces constituting the prism mirror **22** have asymmetry with respect to the longitudinal direction parallel to the YZ plane and intersecting the optical axis AX with the optical axis AX interposed therebetween, and have symmetry with respect to the transverse direction or the X direction with the optical axis AX interposed therebetween. The optical surfaces of the prism mirror **22**, that is, the incident surface **22a**, the inner reflection surface **22b**, and the emission surface **22c** are, for example, free curved surfaces. The incident surface **22a**, the inner reflection surface **22b**, and the emission surface **22c** are not limited to free curved surfaces, and may be aspherical surfaces. The prism mirror **22** may be formed of, for example, a resin, but may also be formed of glass. The inner reflection surface **22b** is not limited to one that reflects the image light ML by total reflection, and may be a reflection surface formed of a metal film or a dielectric multilayer film. In this case, a reflection film formed of a single layer film or multilayer film formed of a metal such as Al or Ag is formed at the inner reflection surface **22b** by vapor deposition or the like, or a sheet-like reflection film formed of a metal is affixed thereto. Although detailed illustration is omitted, an antireflection film can be formed at the incident surface **22a** and the emission surface **22c**.

(35) The see-through mirror **23**, that is, the first combiner **103a**, is a curved plate-like reflecting optical member that functions as a concave surface mirror, reflects the image light ML from the prism mirror **22**, and partially transmits external light OL. The see-through mirror **23** reflects the image light ML from the prism mirror **22** disposed in an emission region of the first projection optical system **12a** toward the pupil position PP. The see-through mirror **23** has a reflection surface **23c** and an outer surface **23o**.

(36) The see-through mirror **23** partially reflects the image light ML and enlarges an intermediate image formed at the light emission side of the emission surface **22c** of the prism mirror **22**. The see-through mirror **23** is a concave mirror that covers the pupil position PP at which the eye EY or the pupil is disposed, has a concave shape toward the pupil position PP, and has a convex shape toward the outside. The pupil position PP or an opening PPa thereof is referred to as eye point or eye box. The pupil position PP or the opening PPa corresponds to an emission pupil EP on the emission side of the first display unit **20a**. The see-through mirror **23** is a collimator and converges main rays of the image light ML temporarily spread by imaging in the vicinity of the emission side of the prism mirror **22** of the first projection optical system **12a**, which are the main rays of the image light ML emitted from each of points on the display surface **11d**, to the pupil position PP. As a concave mirror, the see-through mirror **23** enables an intermediate image (not illustrated) formed at the first display element **11a**, which is an image light generation device, and re-imaged by the first projection optical system **12a** to be seen in an enlarged manner. More specifically, the see-through mirror **23** functions in the same manner as a field lens, and causes the image light ML from each of the points of the intermediate image (not illustrated) formed behind the emission surface **22c** of the prism mirror **22** to be incident on the pupil position PP in a collimated state so as to be collected as a whole. The see-through mirror **23** needs to have a spread equal to or larger than the effective region EA corresponding to an angle of view (the sum of viewing angles in the vertical and horizontal directions with respect to the optical axis AX extending in the forward direction of the eye), from the viewpoint that it is disposed between the intermediate image and the pupil position PP. In the see-through mirror **23**, an outer region extending to the outside of the effective region EA does not directly affect the imaging and thus can have an arbitrary surface shape, but from the viewpoint of ensuring an exterior like a spectacle lens, it is desirable that a curvature of the outer region be the same as a curvature of a surface shape of an outer edge of the effective region EA, or the curvature of the outer region continuously change from the outer edge.

(37) The see-through mirror **23** is a transfective mirror plate having a structure in which a transmissive reflection film **23a** is formed at a rear surface of a plate-like body **23b**. The reflection surface **23c** of the see-through mirror **23** has asymmetry with respect to the longitudinal direction parallel to the YZ plane and intersecting the optical axis AX with the optical axis AX interposed

therebetween, and has symmetry with respect to the transverse direction or the X direction with the optical axis AX interposed therebetween. The reflection surface **23c** of the see-through mirror **23** is, for example, a free curved surface. The reflection surface **23c** is not limited to a free curved surface, and may be an aspherical surface. The reflection surface **23c** needs to have a spread equal to or larger than the effective region EA. When the reflection surface **23c** is formed in the outer region wider than the effective region EA, a difference in visibility is less likely to occur between an external image from behind the effective region EA and an external image from behind the outer region.

(38) The reflection surface **23c** of the see-through mirror **23** transmits some of light when the image light ML is reflected. Thus, because the external light OL passes through the see-through mirror **23**, see-through view of the outside is enabled, and a virtual image can be superimposed on an external image. At this time, when the plate-like body **23b** has a thickness of less than or equal to approximately a few millimeters, a change in magnification of the external image can be curbed to be small. A reflectance of the reflection surface **23c** with respect to the image light ML and the external light OL is set to 10% or more and 50% or less in a range of an incident angle (corresponding to the effective region EA) of the assumed image light ML from the viewpoint of ensuring a brightness of the image light ML and facilitating observation of the external image by see-through. The plate-like body **23b** which is a base material of the see-through mirror **23** is formed of, for example, a resin, and may also be formed of glass. The plate-like body **23b** is formed of the same material as a support plate **61** that supports the plate-like body **23b** from the surrounding thereof, and has substantially the same thickness as the support plate **61**. The transmissive reflection film **23a** is formed of, for example, a dielectric multilayer film configured of a plurality of dielectric layers of which a film thickness has been adjusted. The transmissive reflection film **23a** may be a single-layer film or a multilayer film of a metal such as Al or Ag of which a film thickness has been adjusted. The transmissive reflection film **23a** may be formed by laminating using deposition, for example, and may also be formed by affixing a sheet-like reflection film. An antireflection film is formed at an outer surface **23o** of the plate-like body **23b**.

(39) A light transmission cover **104a** is disposed in front of the see-through mirror **23**. The light transmission cover **104a** is a thin plate-like member having high light transmittance, and an upper end thereof is supported by the cover member **70**, specifically, the front cover **71** illustrated in FIG. 2 and the like. The light transmission cover **104a** has a convex shape toward the outside and has a uniform thickness. The light transmission cover **104a** does not affect the imaging of the image light ML, and a curvature thereof can be arbitrarily set within a range in which it does not interfere with the see-through mirror **23**. The light transmission cover **104a** is as thin as about several millimeters or less and hardly affects observation of an external image. The light transmission cover **104a** is formed of, for example, a resin, and an antireflection film or a hard coating layer may be formed thereon.

(40) In describing the optical path, the image light ML from the first display element **11a** is incident on the projection lens **21** and is emitted from the projection lens **21** in a substantially collimated state. The image light ML that has passed through the projection lens **21** is incident on the prism mirror **22**, passes through the incident surface **22a** while being refracted by it, is reflected by the inner reflection surface **22b** with a high reflectance close to 100%, and is refracted again by the emission surface **22c**. The image light ML from the prism mirror **22** once forms an intermediate image, is then incident on the see-through mirror **23** and is reflected by the reflection surface **23c** with a reflectance of about 50% or less. The image light ML reflected by the see-through mirror **23** is incident on the pupil position PP at which the eye EY or pupil of the wearer US is placed. The external light OL that has passed through the light transmission cover **104a** and has passed through the see-through mirror **23** and the support plate **61** therearound is also incident on the pupil position PP. In other words, the wearer US wearing the first display device **100A** can observe a virtual image of the image light ML in a state in which it overlaps the external image.

(41) The first circuit member **80a** illustrated in FIG. 4 or the second circuit member **80b** illustrated in FIG. 5 includes the display control device **88**. The display control device **88** is a display control circuit, outputs a drive signal corresponding to an image to the first display elements **11a** and the like, and controls display operations of the first display elements **11a** and the like. The display control device **88** includes, for example, an IF circuit, a signal processing circuit, and the like, and causes a two-dimensional image display to be performed on the first display element **11a** and the like according to image data or an image signal received from the outside. The display control device **88** may include a main substrate that controls the first display device **100A** and the second display device **100B**. The main substrate may have an interface function that communicates with the user terminal **90** illustrated in FIG. 1 and performs signal conversion on a signal received from the user terminal **90**, and an integration function that links the display operation of the first display device **100A** and the display operation of the second display device **100B**. The HMD **200** or the image display device **100** that does not include the display control device **88** and the user terminal **90** is also an image display device.

(42) The accessory component circuit member **80c** illustrated in FIG. 4 is operated under the control of the display control device **88**, and operates a camera **3a**, an illuminance sensor **3b**, and a proximity sensor **3c** that are the accessory components illustrated in FIG. 2, and operates a proximity sensor **3d** illustrated in FIG. 3. Specifically, the accessory component circuit member **80c** captures an image of the front of the HMD **200** by the camera **3a**, for example, and acquires a front image. In addition, the accessory component circuit member **80c** detects brightness of the surrounding environment of the HMD **200** by the illuminance sensor **3b**, and outputs it to the display control device **88** as information for controlling a display brightness, for example. The accessory component circuit member **80c** detects an object approaching the HMD **200** from the front by the proximity sensor **3c**, detects an object (specifically, a wearer) approaching the HMD **200** from the rear by the proximity sensor **3d**, and outputs them to the display control device **88**.

(43) FIG. 5 is an exploded perspective view of the HMD **200** or image display device **100**. An optical device **101** is inserted into the middle frame **72** from below, and is fixed to both ends of the middle frame **72** using bearing members **59a**. In the optical device **101**, the first metallic frame **52a** and the second metallic frame **52b** are coupled to each other, the upper side and the side surface thereof are covered with a shield member **87a**, and a joint portion between the first metallic frame **52a** and the second metallic frame **52b** is also covered with the shield member **87a**. The shield member **87a** prevents electromagnetic interference. The first circuit member **80a** is disposed between the first metallic frame **52a** and the shield member **87a**, and the second circuit member **80b** is disposed between the second metallic frame **52b** and the shield member **87a** in a partially exposed state. The first and second circuit members **80a** and **80b** are circuit boards MZ. In the optical device **101**, the circuit boards MZ are sandwiched between the shield member **87a** and the first and second metallic frames **52a** and **52b**, so that waterproofing of the circuit boards MZ can be enhanced. The first circuit member **80a** and the second circuit member **80b** are coupled to each other by a flexible printed circuit (FPC) member **82a** to be electrically communicable with each other. The lower cover **73** is assembled to the lower side of the optical device **101**. A nose pad member **76** is fixed to a central portion of the lower cover **73** from below. The accessory component circuit member **80c** is fixed to an upper portion of the middle frame **72** holding the optical device **101**, and the FPC member **82d** extending from the accessory component circuit member **80c** is coupled to the second circuit member **80b**, a camera device (not illustrated), or the like.

(44) The optical device **101** will be described with reference to FIGS. 6 and 7. FIG. 6 is a perspective view illustrating a state in which the shield member **87a** is fixed by an adhesive, and FIG. 7 is a perspective view illustrating a state in which the shield member **87a** is removed. In the optical device **101**, an optical case CA includes a first optical case CA1 for the right eye and a second optical case CA2 for the left eye.

(45) In the shield member **87a** illustrated in FIG. 6, a bent portion **87h** is provided downward from a corner portion **87f**. The corner portion **87f** of the shield member **87a** is fixed to an upper end of a support structure **50** to be filled with an adhesive on the inner side, thereby sealing a gap between the shield member **87a** and the support structure **50**. The shield member **87a** is formed of a magnesium alloy, for example, and curbs emission of electromagnetic waves from the first circuit member **80a** and the second circuit member **80b** (refer to FIG. 5). Openings **87o** and **87p** are formed in the shield member **87a**, so that an FPC member (not illustrated) coupled to the first circuit member **80a** and the second circuit member **80b** accommodated under the shield member **87a** can be retracted, and heat dissipation from the circuit members **80a** and **80b** accommodated under the shield member **87a** is facilitated. The openings **87o** and **87p** may be covered with a metallic material or a graphite sheet GS having high thermal conductivity and heat dissipation properties after the FPC member is retracted. As a result, it is possible to enhance the sealing property around the circuit members **80a** and **80b**. A peripheral seal member **89a** is mounted on a portion between the first metallic frame **52a** and the second metallic frame **52b** to surround the bent portion **87h** from below and in the forward and rearward direction. The peripheral seal member **89a** seals the periphery of a joint **50c** (refer to FIG. 7), which will be described below, to curb moisture entering an upper portion of the joint **50c** and recessed portions on the metallic frames **52a** and **52b**.

(46) In the first display device **100A** illustrated in FIG. 7, the first metallic frame **52a** is fixed to an upper portion **41t** of the first display unit **20a** illustrated in FIG. 4 using a fastener **50f** such as a screw, and supports the first display unit **20a** in a hanging manner. The first metallic frame **52a** can be fixed by various methods other than screw fixation, such as fixation by caulking, fixation by an adhesive, fixation using fitting, a hanging method such as a socket type, a hanging method by hooking, and a bonding fixation.

(47) A rectangular opening **52o** is formed in the first metallic frame **52a**, and a part of the periphery **52r** of the rectangular opening **52o** (specifically, three sides excluding the +Z side) is in contact with and in close contact with the upper portion **41t** of the barrel **41** of the first display unit **20a**. The first circuit member **80a** is disposed in a recess RE on the first metallic frame **52a**. The first metallic frame **52a** is formed of, for example, a magnesium alloy. When the first metallic frame **52a** is formed of a magnesium alloy, it is possible to provide an effect of cooling the first metallic frame **52a** and the first display unit **20a** by heat dissipation.

(48) In the second display device **100B** illustrated in FIG. 7, similarly to the first display device **100A**, the second metallic frame **52b** is fixed to the upper portion **41t** of the second display unit **20b** using a fastener **50f** such as a screw or the like, and supports the second display unit **20b** in a hanging manner. A rectangular opening **52o** is formed in the second metallic frame **52b**, and a part of the periphery **52r** of the rectangular opening **52o** (specifically, three sides excluding the +Z side) is in contact with and in close contact with the upper portion **41t** of the barrel **41** of the second display unit **20b**. The second circuit member **80b** is disposed in a recess RE on the second metallic frame **52b**. When the second metallic frame **52b** is formed of a magnesium alloy, it is possible to provide an effect of cooling the second metallic frame **52b** and the second display unit **20b** by heat dissipation.

(49) The support structure **50** includes, in addition to the first metallic frame **52a** and the second metallic frame **52b**, a joint **50c** that couples the first metallic frame **52a** and the second metallic frame **52b** to relatively fix them. As in the first metallic frame **52a**, the joint **50c** is a metallic member such as a magnesium alloy, is coupled to one end portion of the first metallic frame **52a** using a fastener **50g** or the like, and is coupled to the other end portion of the second metallic frame **52b** using a fastener **50g** or the like. A fastening portion **50k** for fixing the lower cover **73** is also provided on the joint **50c**. The first metallic frame **52a** in which the first display unit **20a** is mounted and the second metallic frame **52b** in which the second display unit **20b** is mounted are fixed in a mutually optically aligned state via the joint **50c** at the center. Since the two metallic

frames **52a** and **52b** are coupled via the joint **50c** in this way, the arrangement relationship between the two metallic frames **52a** and **52b** can be easily changed. In addition, when other components are incorporated, individual adjustment is possible, and integrated adjustment is also possible. An upper side of the joint **50c** is covered with the shield member **87a**, and a lower side thereof is covered with the peripheral seal member **89a**.

(50) FIG. **8** is a perspective view illustrating a state in which the support structure **50** is removed from the HMD **200** illustrated in FIG. **7**. The first display unit **20a** has the first projection optical system **12a** and the first combiner **103a** in an integrated state, and the second display unit **20b** has the second projection optical system **12b** and the second combiner **103b** in an integrated state. In the first projection optical system **12a**, the first combiner **103a** is fixed to the barrel **41** by adhesion or the like in an aligned state. The barrel **41** of the first projection optical system **12a** has a space for accommodating the first display element **11a**, and supports the first display element **11a** in an aligned state with respect to the projection lens **21** and the like illustrated in FIG. **2**. In the second projection optical system **12b**, the second combiner **103b** is fixed to the barrel **41** by adhesion or the like in an aligned state. The barrel **41** of the second projection optical system **12b** has a space for accommodating the second display element **11b**, and supports the second display element **11b** in an aligned state with respect to the projection lens **21** and the like illustrated in FIG. **2**.

(51) In the first display unit **20a**, a fastening portion **51f** provided in the barrel **41** is, for example, a screw hole, and the upper portion **41t** of the barrel **41** can be fixed to the first metallic frame **52a** by screwing a fastener **50f** illustrated in FIG. **7**. In the second display unit **20b**, the fastening portion **51f** provided in the barrel **41** is, for example, a screw hole, and the upper portion **41t** of the barrel **41** can be fixed to the second metallic frame **52b** by screwing the fastener **50f** illustrated in FIG. **7**.

(52) FIG. **9** is a diagram illustrating the first metallic frame **52a**. In FIG. **9**, a region AR1 is a plan view of the first metallic frame **52a**, and a region AR2 is a perspective view of the back side of the first metallic frame **52a**. The first metallic frame **52a** includes a flat plate portion **55a** having an opening **52o** and having a substantially rectangular shape as a whole, and a pair of reinforcing protrusions **55b** and **55c** protruding upward from the flat plate portion **55a**. The first reinforcing protrusion **55b** disposed on the front side which is the +Z side is provided along the front side of a pair of sides extending in the longitudinal direction of the flat plate portion **55a** in an outer edge of the flat plate portion **55a**. The first reinforcing protrusion **55b** protrudes to the upper side which is the +Y side and the lower side which is the -Y side, and extends in the transverse X direction as a whole while being slightly curved. In addition, the second reinforcing protrusion **55c** disposed on the rear side which is the -Z side is provided along the rear side of the pair of sides extending in the longitudinal direction of the flat plate portion **55a** in the outer edge of the flat plate portion **55a**. The second reinforcing protrusion **55c** protrudes only to the upper side which is the +Y side, and extends in the transverse X direction as a whole while being slightly curved. The pair of reinforcing protrusions **55b** and **55c** increase a structural strength of the first metallic frame **52a**, and increase a strength of the flat plate portion **55a** which is relatively weak against bending and twisting.

(53) A planar portion **56t** and a cooling fin structure **56s** are formed at a front surface of the reinforcing protrusion **55b**, that is, on a front portion **55f** of the reinforcing protrusion **55b**. A lead-out end **11q** which is the other end of a heat dissipation sheet **11s** described below is attached to the planar portion **56t** (refer to FIG. **7**). Due to the lead-out end **11q** of the heat dissipation sheet **11s** being attached to the front portion **55f** away from the face of the wearer US, there is an effect of relatively lowering a temperature of a rear portion of the first metallic frame **52a**, and thus a temperature of the HMD **200** or the rear side of the image display device **100** can be relatively lowered. A cooling fin structure **56s** has a three dimensional shape in which ridges and grooves are alternately formed, and increases a contact area with air. The cooling fin structure **56s** is disposed in the vicinity of the left and right sides of the planar portion **56t**. The ridges and grooves constituting the cooling fin structure **56s** extend in the longitudinal Y direction. In the cooling fin structure **56s**, heat received by the planar portion **56t** and diffused along the reinforcing protrusion

55b is radiated from the surfaces through air or radiated as infrared rays. Due to the cooling fin structure **56s** being formed into a pattern shape extending in the longitudinal direction, a convective flow in the longitudinal direction can be easily generated, and it is possible to enhance the effect of heat dissipation to the air. The arrangement and structure of the planar portion **56t** and the cooling fin structure **56s** are merely examples, and the planar portion **56t** and the cooling fin structure **56s** can be provided in a place other than the reinforcing protrusion **55b**.

(54) In the first metallic frame **52a**, a hole **56a** formed in a side portion **55p** at one end is used to couple the first metallic frame **52a** with the joint **50c**. In the first metallic frame **52a**, a hole **56j** formed in a side portion **55q** at the other end is used to couple the first metallic frame **52a** with the middle frame **72** (refer to FIG. 5) via a bearing member **59a**. In the first metallic frame **52a**, holes **56b** formed at four locations around the opening **52o** are used to fix the first metallic frame **52a** to the upper portion **41t** of the barrel **41**. That is, the first metallic frame **52a** can be stably fixed to the upper portion **41t** of the barrel **41** by screwing the fastener **50f** illustrated in FIG. 7 into the fastening portion **51f** illustrated in FIG. 8 through the hole **56b**.

(55) In the first metallic frame **52a**, a space above the flat plate portion **55a** and sandwiched between the pair of reinforcing protrusions **55b** and **55c** is the recess RE for accommodating the first circuit member **80a**. A height of an upper end of the first circuit member **80a** may be higher than that of upper ends of the pair of reinforcing protrusions **55b** and **55c**. In the first metallic frame **52a**, screw holes **56c** formed at three locations in the periphery **52r** of the opening **52o** are used to fix the first circuit member **80a** to the first metallic frame **52a**.

(56) Although not illustrated in the drawings, the second metallic frame **52b** has a shape and structure inverted with respect to the left and right direction, that is, the $\pm X$ direction, of the first metallic frame **52a**. The first metallic frame **52a** itself may have a bilaterally symmetrical shape, and in this case, the second metallic frame **52b** has the same shape as the first metallic frame **52a** without being inverted.

(57) A contour shape of the flat plate portion **55a** of the first metallic frame **52a** does not need to be a rectangular shape, and can be appropriately changed in accordance with the shape and application of the barrel **41**. The shape of the opening **52o** does not need to be rectangular. The pair of reinforcing protrusions **55b** and **55c** are not limited to those extending along the longitudinal side of the flat plate portion **55a**, and may extend along a part of the longitudinal side of the flat plate portion **55a**, may extend along a side other than the longitudinal side, or may be provided in a rib shape inside a portion other than the side.

(58) FIG. 10 is a perspective view illustrating a modified example of the support structure **50** illustrated in FIG. 7. In the optical device **101** illustrated in FIG. 10, the optical case CA includes a first optical case CA1 for the right eye and a second optical case CA2 for the left eye. The support structure **150** illustrated in FIG. 10 is a member in which the first metallic frame **52a**, the second metallic frame **52b**, and the joint **50c** are integrally formed. That is, the support structure **150** including the first and second metallic frames **52a** and **52b** is an integrated member that supports the first optical case CA1 and the second optical case CA2. Thus, shapes of the first and second metallic frames **52a** and **52b** can be simplified. A bent portion **87h** provided on the outer periphery of the shield member **87a** illustrated in FIG. 6 is not present on the outer periphery of a shield member **187a** of the modified example.

(59) FIG. 11 is a plan view of components of the cover member **70** constituting the exterior. In FIG. 11, a region BR1 is a plan view of the front cover **71**, a region BR2 is a plan view of the middle frame **72**, and a region BR3 is a plan view of the lower cover **73**. The front cover **71** is formed of a metal such as SUS, but may be formed of a resin material. The middle frame **72** is formed of a resin material, but may be formed of a metal. The lower cover **73** is formed of a metal such as SUS, but may be formed of a resin material.

(60) Referring also to FIG. 5, the front cover **71** has a top plate **71a** and a front plate **71b**. The top plate **71a** is wide in the forward and rearward direction at the center and gradually decreases in the

forward and rearward width at both left and right ends. A vertical width of the front plate **71b** increases at both left and right ends. At the center of the front plate **71b**, there is formed an opening **78a** for exposing accessory components such as a camera. The front cover **71** has a uniform thickness, and a contour of the front cover **71** in plan view is substantially the same as that of the middle frame **72**.

(61) Referring also to FIG. 5, the middle frame **72** has a front plate **72a**, a rear frame **72b**, a rear plate **72c**, an end portion **72d**, and a lower plate region **72f**. The front plate **72a** has a substantially uniform vertical width. At the center of the front plate **72a**, a plurality of storage chambers **77a** for storing accessory components such as a camera are provided to protrude from a back surface to the $-Z$ side. The upper end side of the front plate **72a** is in contact with or fitted to the front plate **71b** of the front cover **71**. The rear frame **72b** is provided on the upper end side and is in contact with or fitted to a rear edge of the top plate **71a** of the front cover **71**. Although details will be described below, a first seal member **SW1** is provided at a first facing portion **FC1** at which the front cover **71** and the middle frame **72** are in contact with or fitted to each other (refer to FIG. 20). The first facing portion **FC1** is disposed to make substantially one round on a side surface of the front cover **71** or the middle frame **72**, and the first seal member **SW1** is waterproofed or dustproofed on an inner side surface of the exterior case **7**. A pair of pin-like protrusions **72i** formed at the rear frame **72b** are for fixing the accessory component circuit member **80c** illustrated in FIG. 5. The center of the front plate **72a** and the center of the rear frame **72b** are coupled by a bridge **72p**. The rear plate **72c** is provided to protrude downward, that is, to the $-Y$ side from the rear frame **72b**. The rear plate **72c** has a notch **72j** at two locations, and a vertical width thereof is reduced at the notch **72j**. An end portion **72d** is coupled to both ends of the front plate **72a**, and is coupled to both ends of the rear frame **72b** or the rear plate **72c**. A recessed portion **77c** for accommodating electronic components (not illustrated) is formed in the end portion **72d**. A fastening portion **72m** for coupling with the front cover **71** and the lower cover **73** and a fastening portion **72n** for coupling with the lower cover **73** are provided at the end portion **72d**. Further, the bearing member **59a** of the optical device **101** is fixed to the lower surface side by the fastening portion **72n**. The lower plate region **72f** extends rearward, that is, to the $-Z$ side, from a lower end of the central portion of the front plate **72a**. Two openings **AP1** are formed in the middle frame **72** in plan view, and the first metallic frame **52a** and the second metallic frame **52b** of the support structure **50** can be accommodated therein.

(62) Referring also to FIG. 5, the lower cover **73** has a central portion **73a**, a barrel accommodating portion **73b**, and an end portion **73c**. The central portion **73a** has a bottom wall **73h** and a rear wall **73i**. A protruding region **73d** is formed at the bottom wall **73h**. The protruding region **73d** and a side portion on the root side thereof are disposed to partially overlap the lower plate region **72f** of the middle frame **72**, and are in contact with or fitted to the lower plate region **72f**. A fastening portion **73p** for fixing the lower cover **73** to the lower plate region **72f** of the middle frame **72** is provided at the protruding region **73d**. A fastening portion **73q** for fixing the nose pad member **76** and the joint **50c** to the lower cover **73** is provided at a main body of the bottom wall **73h**. The rear wall **73i** is disposed near the outer side of the rear plate **72c** of the middle frame **72**, or is in contact with or fitted to the outer side of the rear plate **72c**. The rear wall **73i** is also formed at the end portion **73c**. A pair of barrel accommodating portions **73b** are provided to sandwich the central portion **73a**, and have a downward protruding portion **73s** and a rearward protruding portion **73t**. The barrel accommodating portion **73b** is a portion that covers the barrel **41** of the first projection optical system **12a** or the second projection optical system **12b**, and has an opening **AP2**. The opening **AP2** is provided corresponding to the emission port **41o** of the barrel **41**, and allows the image light to pass therethrough. An upper end **73k** of the rearward protruding portion **73t** is in contact with or fitted to the rear frame **72b** of the middle frame **72**. Although details will be described below, a second seal member **SW2** is provided at a second facing portion **FC2** at which the middle frame **72** and the lower cover **73** are in contact with or fitted to each other (refer to FIG.

20). The second facing portion FC2 is waterproofed or dustproofed at a joint surface on the rear side of the exterior case 7. A fastening portion 73m for fixing the lower cover 73 to the fastening portion 72m at the top plate 71a of the front cover 71 or the end portion 72d of the middle frame 72, and a fastening portion 73n for fixing the lower cover 73 to the fastening portion 72n at the end portion 72d of the middle frame 72 are provided at the end portion 73c.

(63) FIG. 12 is a side view of the HMD 200 or the image display device 100, and is a diagram illustrating an angle change of the optical device 101. In FIG. 12, a region CR1 indicates a first state of the optical device 101, and a region CR2 indicates a second state of the optical device 101. The optical device 101 is supported by the bearing member 59a to be rotatable about a rotation axis RX extending in parallel with the X-axis, and can change an angular posture thereof with an upper limit of about 5 to 10°. In this case, an angle of the optical device 101 with respect to the cover member 70 or the middle frame 72 changes, in the first state, the combiners 103a and 103b are separated from the light transmission covers 104a and 104b, and in the second state, the combiners 103a and 103b are relatively close to the light transmission covers 104a and 104b. In the case of the first state, a virtual image is observed downward on the assumption that a position of the eye is located on the relatively upper side. In the case of the second state, the virtual image is observed on the relatively upper side on the assumption that the position of the eye is located on the relatively lower side. The visibility can be improved by allowing the rotation of the combiner 103.

(64) FIG. 13 is a partial perspective view of the back surface side of the HMD 200 or image display device 100. The optical device 101 is movable with respect to the middle frame 72 and the lower cover 73, and has gaps GP3, GP4, and GP5 between the middle frame 72 and the lower cover 73. More specifically, the gap GP5 is formed between the front side of the upper end 61g of each of the combiners 103a and 103b and a rear end 72x of the lower plate region 72f of the middle frame 72. Further, the gaps GP3 and GP4 are formed between the upper ends 61g of each of the combiners 103a and 103b and a front end 73x of the downward protruding portion 73s of the lower cover 73. Although the details will be described below, it is desirable that the gaps GP3, GP4, and GP5 are set to achieve dustproof and waterproof while considering a size of a flow path to the inside of the cover member 70 and a method of bending a flow path to the inside of the cover member 70.

(65) The structure of the barrel 41 incorporated in the optical device 101 illustrated in FIG. 7 and the like will be described with reference to FIG. 14. In FIG. 14, a region DR1 is a side cross-sectional view of the barrel 41 and the first display element 11a and the optical members 2a and 2b held by the barrel 41, a region DR2 is a side cross-sectional view of a state in which the first display element 11a and a holder 31 are removed, and a region DR3 is a side cross-sectional view of a state in which a barrel cover 41u is further removed.

(66) The barrel 41 includes a barrel main body 41a and the barrel cover 41u, accommodates a first optical member 2a, and holds a second optical member 2b. The barrel main body 41a and the barrel cover 41u are formed of a polycarbonate resin in consideration of support accuracy and strength of the optical elements fixed therein. The barrel main body 41a is a bathtub-like vessel of which an upper portion is open, and has the emission port 41o at a part of the bottom. The barrel cover 41u is fixed to cover the barrel main body 41a from above. The barrel main body 41a includes two side plate members 41c, a bottom plate member 41d, and a front plate member 41e. The two side plate members 41c extend substantially parallel to an off-axis plane (parallel to the YZ plane) in which the optical axis AX extends and are spaced apart from each other. The bottom plate member 41d extends substantially along the XZ plane perpendicular to the off-axis plane (parallel to the YZ plane) in which the optical axis AX extends, and the emission port 41o is provided on the rear end side. The front plate member 41e couples a front end of the bottom plate member 41d to front ends of the two side plate members 41c.

(67) As illustrated in FIG. 8, the barrel main body 41a has two protruding portions 41f and 41g extending in the transverse direction or the $\pm X$ direction to protrude outward from upper portions

of the two side plate members **41c**. The protruding portions **41f** and **41g** are portions for fixing the upper end **61g** of the first combiner **103a**, and are disposed to face a back surface of the reinforcing protrusion **55b** of the first metallic frame **52a** illustrated in FIG. 7.

(68) Referring back to FIG. 14, on the inner side of one of the side plate members **41c**, there are formed guide convex portions **45a**, **45b**, **45c**, and **45d** having steps, as protrusions for supporting a first lens **21o**, a second lens **21p**, and the third lens **21q** which constitute the first optical member **2a**, and the prism mirror **22** of the second optical member **2b**. Although not illustrated in the drawings, guide protrusions similar to the guide convex portions **45a**, **45b**, **45c**, and **45d** are formed at an inner surface of the other side plate member **41c** (refer to FIG. 15). The first lens **21o** is positioned in a biased state by the two first guide convex portions **45a** provided on the inner surfaces of the two side plate members **41c** and is supported by the barrel main body **41a**. Similarly, the second lens **21p** is positioned by the second guide convex portion **45b** and supported by the barrel main body **41a**, the third lens **21q** is positioned by the third guide convex portion **45c** and supported by the barrel main body **41a**, and the prism mirror **22** is positioned by the fourth guide convex portion **45d** and supported by the barrel main body **41a**. When the optical elements **21o**, **21p**, **21q**, and **22** are positioned by the guide convex portions **45a**, **45b**, **45c**, and **45d**, positioning surfaces of the optical elements **21o**, **21p**, **21q**, and **22** are brought into contact with a plurality of positioning surfaces formed at the guide convex portions **45a**, **45b**, **45c**, and **45d**, and contact portions or peripheries thereof are fixed with an adhesive. A method for fixing the prism mirror **22** and the like to the barrel main body **41a** is not limited to the above-described method using the bias at the contact surface, but may be replaced with a method using fitting or other various methods.

(69) The barrel cover **41u** is disposed on the opposite side of the bottom plate member **41d** and covers the inside of the barrel main body **41a** to form a storage space **IS**. The barrel cover **41u** includes a top plate **41x** and a rear plate **41y**. The top plate **41x** extends parallel to the XZ plane, and the rear plate **41y** is disposed to be inclined so as to cover the outer side of the inner reflection surface **22b** of the prism mirror **22** of the second optical member **2b**. In the barrel cover **41u**, a positioning holder pedestal **41s** lowered by a predetermined height from the periphery is formed at the front +Z side, and an insertion port **41z** is formed in front of the holder pedestal **41s**. The holder pedestal **41s** provided on the barrel cover **41u** faces a base plate **31b** of the holder **31**, which will be described below, at the time of and after assembly as described below. The base plate **31b** is fixed to the barrel **41** while covering the insertion port **41z**. An inner surface **41m** of the rear plate **41y** is inclined with respect to the XZ plane and the XY plane, and extends along the inner reflection surface **22b** of the prism mirror **22** to the vicinity of the inner reflection surface **22b**. A uniform gap **GA** is formed between the outer side of the inner reflection surface **22b** and the inner surface **41m** of the rear plate **41y**.

(70) As illustrated in FIG. 15, fitting structures **47a** and **47b** such as steps are provided between an outer edge **42q** extending along an outer periphery of the barrel cover **41u** and an upper end **42p** of the barrel main body **41a** to achieve mutual positioning. The outer edge **42q** of the barrel cover **41u** and the upper end **42p** of the barrel main body **41a** form a coupling portion **CJ** between the barrel main body **41a** and the barrel cover **41u**. In the coupling portion **CJ**, a gap between the outer edge **42q** of the barrel cover **41u** and the upper end **42p** of the barrel main body **41a**, that is, a gap between the fitting structures **47a** and **47b** and the outer edge **42q** or the upper end **42p** is filled with a seal member **SM** which functions as an adhesive material or a sealing material (refer to a region **DR2** in FIG. 14). In this case, airtightness of the storage space **IS** can be enhanced. The seal member **SM** provided along the outer edge **42q** of the barrel cover **41u** serves to seal the barrel **41** together with the seal member **SM** (described below) provided around the insertion port **41z** and the emission port **41o**. The waterproof property of the image display device **100** can be improved by sealing the barrel **41** which is the optical case **CA**.

(71) Referring to FIG. 14, a diaphragm plate member **26** is disposed in the barrel **41** between the

first optical member **2a** and the second optical member **2b**. In the illustrated case, the diaphragm plate member **26** is mounted adjacent to the incident surface **22a** of the prism mirror **22**.

(72) A gap between an outer edge of the emission surface **22c** of the prism mirror **22** and an edge portion **44** of the emission port **41o** on the inner side is filled with a seal member SM that functions as an adhesive or a sealing material. The seal member SM seals between the emission port **41o** of the barrel main body **41a** and the periphery of the second optical member **2b** or the emission surface **22c** of the prism mirror **22**. In this case, the emission surface **22c** of the second optical member **2b** is exposed to the outside, but an optical surface optically upstream of the emission surface **22c** of the second optical member **2b** is protected by the dustproof and waterproof structure of the barrel **41**. The seal member SM filled along the emission port **41o** of the barrel main body **41a** is an elastic adhesive AO. The elastic adhesive AO is, for example, a silicone-based photo-curable resin that is cured by curing light such as UV light, but has elasticity even after curing. The elastic adhesive AO enables the emission port **41o** to be dustproofed and waterproofed.

(73) In the barrel **41**, the first display element **11a** supported by the holder **31** is inserted into a space ISa facing the front plate member **41e** from above through the insertion port **41z** and fixed in a positioned state. The barrel **41** is sealed by the holder **31** or the like. In this case, since the first display element **11a** is disposed in the barrel **41**, the first display element **11a** is less likely to be affected by an impact from the outside, and a situation in which misalignment occurs due to an operation error in a manufacturing process is less likely to occur. The first display element **11a** is disposed by the holder **31** at the front end of the barrel **41**, that is, the optical case CA in the +Z direction so that the display surface **11d** faces rearward corresponding to the -Z direction.

(74) FIG. **16** is a diagram illustrating a display unit DU in which the first display element **11a** is assembled to the holder **31**. In FIG. **16**, a region ER1 is a perspective view illustrating the front side of the display unit DU, a region ER2 is a side cross section of the display unit DU, and a region ER3 is a perspective view illustrating the back side of the holder **31**.

(75) In the illustrated display unit DU, the first display element **11a** and a light shielding plate **33** associated therewith are fixed to a holder **31** and are positioned relative to each other.

(76) The first display element **11a** has a plate-like main body portion **11k** and a flexible printed circuit (FPC) portion **11f** that is coupled to an upper portion of the main body portion **11k** and extends upward. In these portions, the main body portion **11k** includes a silicon substrate SS in which a driving circuit **11j** is formed and forming an exterior of the main body portion **11k**, a light emission layer **11e** which is an organic EL element containing an organic EL material and generates light of a color needed for image light ML, and a protective glass GG for sealing that seals the light emission layer **11e** in cooperation with the silicon substrate SS. Here, the light emission layer **11e** corresponds to the display surface **11d**. The first display element **11a** emits the image light ML to the protective glass GG side by performing a light emission operation in accordance with a driving signal received from the FPC portion **11f**. An introduction end **11p** which is one end of a flexible or elastic heat dissipation sheet **11s** is attached to on the back surface SSa of the silicon substrate SS. The heat dissipation sheet **11s** serves to conduct heat from the first display element **11a** to the first metallic frame **52a** illustrated in FIG. **7**. The heat dissipation sheet **11s** is a heat dissipation member RM formed of, for example, graphite and is bonded to a back surface SSa of the silicon substrate SS using an adhesive having high thermal conductivity.

(77) The holder **31** is a member formed of, for example, a resin having a light shielding property, and has an exterior bent in an L-shape in side view. The holder **31** includes a support frame **31a** that supports the first display element **11a**, and a base plate **31b** that is coupled to an upper portion of the support frame **31a** and extends in a direction intersecting (for example, orthogonal to) the support frame **31a**. The support frame **31a** is inserted into the barrel **41** through the insertion port **41z** formed in the barrel **41** while supporting the first display element **11a** (refer to FIG. **14**). The base plate **31b** is coupled to the root side of the support frame **31a**, extends forward (that is, to the -Z side) corresponding to the light emission side, and is not inserted into the barrel **41**. The support

frame **31a** has a rectangular exterior and includes a flat plate portion **31s** and a frame portion **31t**. An upper end of the flat plate portion **31s** is coupled to the base plate **31b**. The frame portion **31t** has a U shape and surrounds the first display element **11a** in the left and right direction and a lower direction. The support frame **31a** has a rectangular opening **A1** surrounded by the flat plate portion **31s** and the frame portion **31t**. The protective glass **GG** of the first display element **11a** is disposed to be fitted into the opening **A1**. Inside the support frame **31a**, two support regions **31p** extending parallel to the transverse **X** direction are formed in an upper portion and a lower portion in the **Y** direction. The upper support region **31p** is formed as a ridge on the back surface side of the flat plate portion **31s**, and the lower support region **31p** is formed as a step on the back surface side of the frame portion **31t**. The both support regions **31p** are bonded to the upper and lower surface regions **SSc** in the silicon substrate **SS** of the first display element **11a** via an adhesive. As a result, the first display element **11a** is supported in a state in which it is indirectly positioned with respect to the support frame **31a**, and the display surface **11d** of the first display element **11a** can be brought into a predetermined state in which it is positioned substantially parallel to the **XY** plane. The base plate **31b** of the holder **31** has a rectangular flat plate exterior, and a lower surface **31j** extends in parallel with the **XZ** plane. The base plate **31b** is placed on the holder pedestal **41s** formed at the barrel cover **41u** of the barrel **41**, and is fixed to the holder pedestal **41s** after positioning (refer to FIG. **14** and the like). In an outer peripheral portion of the base plate **31b**, thin portions **35t** are formed at three sides of the rear side, that is, the $-Z$ side, and the transverse side, that is, the $\pm X$ side. An upper surface **31u** of the base plate **31b** is smooth and planar in order to facilitate suction and support by an arm of a three dimensional driving device for positioning.

(78) The light shielding plate **33** is fixed to the support frame **31a** of the holder **31** using an adhesive or a cohesive material. The light shielding plate **33** is a light shielding diaphragm having a rectangular opening **33p**, and is formed of a metal, a resin, or the like having a light shielding property. The effective image light **ML** emitted from the display surface **11d** of the first display element **11a** passes through the opening **33p** without being blocked by the light shielding plate **33**. When the light shielding plate **33** is fixed, positioning is performed so that four protrusions **31q** formed at the support frame **31a** grip protruding portions **33c** formed at the left and right sides of the light shielding plate **33** from above and below.

(79) The fixing of the display unit **DU** to the barrel **41** will be described with reference to FIGS. **17** and **18**. FIG. **17** is an enlarged cross-sectional view illustrating a front portion of the barrel **41**, and FIG. **18** is a view illustrating assembly of the optical unit **300**. In FIG. **18**, a region **FR1** is a plan view illustrating a state in which the display unit **DU** is assembled to the barrel **41**, and a region **FR2** is a plan view illustrating a state before the display unit **DU** is assembled. Here, an assembly in which the display unit **DU** including the first display element **11a** is assembled to the first display unit **20a** in which the barrel **41** incorporating the first projection optical system **12a** and the first combiner **103a** are combined is referred to as an optical unit **300**.

(80) A step **S1** is formed in the left, right, and rear edge portions **41r** of the holder pedestal **41s** formed at the top plate **41x** of the barrel cover **41u** which is an upper surface of the barrel **41**. The step **S1** of the holder pedestal **41s** and the vicinity thereof hold an adhesive **AM1** that couples the holder **31** and the barrel **41** together. The adhesive **AM1** is, for example, an acrylic ultraviolet curable resin. The adhesive **AM1** is cured after the holder **31** is positioned as described below.

(81) In the holder **31** of the display unit **DU**, when a lower end of the support frame **31a** is inserted from the insertion port **41z** and the entire support frame **31a** advances into the barrel **41** together with the first display element **11a**, the first display element **11a** is accommodated in a space **ISa**, the base plate **31b** is placed to be fitted into the recessed holder pedestal **41s**, and a seat surface **41n** and a lower surface **31j** closely face each other. At this time, most of the insertion port **41z** is closed by the base plate **31b**, and thus dust and dirt are prevented from entering the holder **31**.

(82) The base plate **31b** of the holder **31** has a thin portion **35t** at left, right and rear edge portions. The thin portion **35t** forms a step **S2** facing the step **S1** of the holder pedestal **41s**. As a result, a

groove TR is formed by the step S1 provided in the holder pedestal 41s of the barrel cover 41u and the step S2 provided in the thin portion 35t of the holder 31. The groove TR is a part of an adhesive application portion AA, and serves to hold the adhesive AM1 supplied between the holder 31 and the barrel 41 around the thin portion 35t of the base plate 31b.

(83) A size of the base plate 31b of the holder 31 in plan view, that is, a size projected onto the XZ plane is smaller than a size of the holder pedestal 41s in plan view and is smaller than a size of the opening 52o formed in the first metallic frame 52a illustrated in FIG. 7 in plan view. Thus, even in a state in which the first metallic frame 52a is fixed to the barrel 41, the holder 31 can be placed on the upper portion 41t of the barrel 41 over the opening 52o of the first metallic frame 52a.

(84) In order that a position of the support frame 31a of the holder 31 can be slightly moved in the X direction and the Z direction within the insertion port 41z, the outline of the insertion port 41z in plan view is made slightly larger than the outlines of the support frame 31a and the first display element 11a in plan view. As a result, the insertion port 41z is partially opened to form a gap G1 in the front upper portion of the holder 31, that is, on the front +Z side of a joint portion between the base plate 31b and the support frame 31a. That is, the insertion port 41z has a gap G1 between the insertion port 41z and the holder 31 on the rear surface side or the +Z side of the first display element 11a. The FPC portion 11f which is a wire 11r extending from the first display elements 11a, and the heat dissipation sheet 11s extend to the outside of the barrel 41 via the gap G1. Before the holder 31 is positioned and fixed with respect to the barrel 41, an adhesive AM2 serving as a sealing portion is applied to be filled between a front end of the insertion port 41z and the heat dissipation sheet 11s of the first display element 11a, the FPC portion 11f, and the like to cover the gap G1, and the adhesive AM2 is cured after the holder 31 is positioned as described below. The adhesive AM2 has a relatively high viscosity before being cured so that the shape thereof can be easily maintained. When the adhesive AM1 corresponding to the groove TR and the adhesive AM2 corresponding to the gap G1 are combined, a shape closed like four sides of a rectangle is obtained. A portion (the groove TR or the gap G1) to which the adhesives AM1 and AM2 serving as the seal member SM are applied becomes an adhesive application portion AA.

(85) Fixing of the first combiner 103a to the barrel 41, that is, fixing of the see-through mirror 23 to the first projection optical system 12a will be described with reference to FIG. 19. In FIG. 19, a region GR1 is a front view of the barrel 41 and the first combiner 103a, and a region GR2 is a plan view of the barrel 41 and the first combiner 103a.

(86) In the optical unit 300, a pair of protruding portions 41f and 41g are formed at the front side of the barrel 41 to protrude outward in the transverse direction. Further, a pair of mounting portions 62a and 62b are formed at an upper end 61g of the first combiner 103a to protrude inward, that is, to the -Z side. A pair of facing inner side surfaces 62s of the pair of mounting portions 62a and 62b are fitted to a pair of outward transverse side surfaces 51s of the barrel 41 to sandwich the pair of transverse side surfaces 51s, and positioning in the $\pm X$ direction is performed to reduce inclination. A pair of rear side surfaces 62t of the pair of mounting portions 62a and 62b are in contact with a pair of stepped front side surfaces 51r of the barrel 41, and positioning in the $\pm Z$ direction is performed to reduce inclination. Further, a plurality of convex portions 59p protruding from bottom surfaces 59j of the pair of protruding portions 41f and 41g are in contact with a pair of upper surfaces 62j of the pair of mounting portions 62a and 62b, and positioning in the $\pm Y$ direction is performed. After the above-described positioning, that is, after the positioning of six axes, an adhesive AM5 is supplied from the periphery between the bottom surfaces 59j of the protruding portions 41f and 41g and the upper surfaces 62j of the mounting portions 62a and 62b, the supplied adhesive AM5 is cured by ultraviolet rays or the like, and thus the fixing of the first combiner 103a to the barrel 41 is completed.

(87) Waterproofing or dustproofing in the image display device 200 will be described with reference to FIGS. 20 and 21. FIG. 20 is a cross-sectional view taken along line A-A of the HMD 200 or image display device 100 illustrated in FIG. 2, and FIG. 21 is a cross-sectional view taken

along line B-B of the HMD **200** or image display device **100** illustrated in FIG. 2.

(88) The image display device **200** includes a first seal member **SW1** in a first facing portion **FC1** between the upper case **CV1** and the lower case **CV2**. Specifically, a gap **GP1** is formed between the front plate **71b** of the front cover **71** and the front plate **72a** of the middle frame **72**, and the gap **GP1** is sealed by arranging the first seal member **SW1** formed of, for example, an elastically deformable resin material to be sandwiched between a back surface of the front plate **71b** and a front surface of the front plate **72a**. The first seal member **SW1** can be fixed to at least one of the back surface of the front plate **71b** and the front surface of the front plate **72a** with an adhesive. Similarly, although a gap **GP1** is formed between the rear end of the top plate **71a** of the front cover **71** and the rear frame **72b** of the middle frame **72**, the gap **GP1** is sealed by arranging the first seal member **SW1** to be sandwiched between a lower surface of the rear end of the top plate **71a** and an upper surface of the rear frame **72b**. The first seal member **SW1** can be fixed to at least one of the lower surface of the rear end of the top plate **71a** and the upper surface of the rear frame **72b** with an adhesive. The first seal member **SW1** is formed in a closed ring shape along the upper end of the middle frame **72**, and achieves sealing over the entire circumference (refer to the first facing portion **FC1** in FIG. 11). Thus, moisture **WA1** entering between the combiner **103** and the shade **104** is prevented from passing through the first facing portion **FC1**.

(89) The image display device **200** includes a second seal member **SW2** in a second facing portion **FC2** between the lower cover **73** and the middle frame **72**. Specifically, a gap **GP2** is formed between an upper end or the rear wall **73i** of the rear protruding portion **73t** of the lower cover **73** and the rear frame **72b** or the rear plate **72c** of the middle frame **72**, and the gap **GP2** is sealed by arranging the second seal member **SW2** formed of, for example, an elastically deformable resin material to be sandwiched between the upper end or the rear wall **73i** of the rear protruding portion **73t** of the lower cover **73** and the rear frame **72b** or the rear plate **72c** of the middle frame **72**. The second seal member **SW2** can be fixed to at least one of the lower cover **73** and the middle frame **72** with an adhesive. The second seal member **SW2** is formed along the rear frame **72b** of the middle frame **72** and terminates at the end portion **72d** (refer to the second facing portion **FC2** in FIG. 11).

(90) As illustrated in FIG. 21, the image display device **200** has a gap **GP3** as a partially bent or narrowed first gap between the combiner **103** and the lower case **CV2**. Specifically, the gap **GP3** which is a first gap is generated in a third facing portion **FC3** between the upper end **61g** of the combiner **103** and the front end of the bottom wall **73h** or the end portion **73c** of the lower cover **73**, and the gap **GP3** serves as a flow path. However, since the gap **GP3** which is the first gap has a curved portion **BE** which is bent and narrowed at the upper end, the moisture **WA3** which enters between the combiner **103** and the shade **104** is curbed from passing through the gap **GP3**.

(91) As illustrated in FIG. 20 and the like, the image display device **200** has a gap **GP4** as a second gap which is partially bent or narrowed between the barrel **41** which is the optical case **CA** and the lower case **CV2**.

(92) Specifically, a gap **GP4a** which is a second gap is generated in a fourth facing portion **FC4a** between the rear surface of the barrel **41** and the rear protruding portion **73t** of the lower cover **73**, and the gap **GP4a** serves as a flow path. However, the lower case **CV2** is bent along the exterior of the optical case **CA** on the pupil position **PP** side (refer to FIG. 4). As a result, the gap **GP4a** has a curved portion **BE** bent along the exterior of the optical case **CA** on the lower end side. In order to accommodate the prism mirror **22**, the optical case **CA** has a protrusion on the lower portion of the rear side which is the pupil position **PP** side. The gap **GP4a** which is the second gap is bent along the protrusion of the optical case **CA**. Due to the bent gap **GP4a**, the moisture **WA4** which enters between the barrel **41** that is the optical case **CA** and the lower case **CV2** is curbed from passing through the gap **GP4a**.

(93) A gap **GP4b** which is a second gap is generated in a fourth facing portion **FC4b** between the lower surface of the barrel **41** and the downward protruding portion **73s** of the lower cover **73**, and

the gap GP4b serves as a flow path. In the fourth facing portion FC4b, since the gap GP4b which is the second gap extends straight, the moisture WA4 that enters between the barrel 41 and the lower case CV2 passes through the gap GP4b but does not go around onto the barrel 41.

(94) The image display device 200 has a gap GP5 as a third gap which is partially bent or narrowed between the combiner 103 and the lower case CV2. Specifically, the gap GP5 which is a third gap is formed in a fifth facing portion FC5 between the lower end of the front plate 72a of the middle frame 72 and the upper end 61g of the combiner 103. However, since the gap GP5 is narrower than a gap GP0 between the combiner 103 and the shade 104, the moisture WA1 that enters between the combiner 103 and the shade 104 is curbed from passing through the gap GP5.

(95) In the above description, the gap GP3 which is the first gap, the gap GP4 which is the second gap, and the gap GP5 which is the third gap allow the rotation of the optical case CA and the combiner 103. In the present embodiment, the rotation of the optical device 101 is ensured from the viewpoint of improving visibility, and it is necessary to increase the gaps GP3, GP4, and GP5 to some extent. Even when the gaps GP3, GP4, and GP5 are provided, the gaps GP3, GP4a, and GP5 that are partially bent or narrowed make it difficult for water to enter the inside. Thus, in the image display device 100, it is possible to achieve both the waterproof structure and the improvement in visibility. Each of the gaps GP3, GP4a, and GP5 has a sufficient length in an entering direction to prevent moisture from entering the circuit board MZ. Clearance values of the gaps GP3, GP4a, and GP5 and widths in a direction orthogonal to a plane in which the gaps GP3, GP4a, and GP5 extend are, for example, 2 to 3 mm, and in particular, regarding the gaps GP3 and GP5, the clearance values are about 1 to 2 mm or less at a narrowing portion.

(96) FIG. 22 is a cross-sectional perspective view illustrating an image display device 100 of the modified example. The image display device 200 includes a third seal member SW5 between the lower case CV2 and the combiner 103. Specifically, the gap GP5 is sealed by arranging the third seal member SW5 formed of an elastic body such as foamed urethane to be sandwiched between the lower end of the front plate 72a of the middle frame 72 and the upper end 61g of the second combiner 103b. The third seal member SW5 is fixed to one of the second combiner 103b and the middle frame 72 with an adhesive. In the third seal member SW5, a surface that is not fixed is smooth, and allows displacement or movement of the second combiner 103b or the middle frame 72 that faces the surface. Since the third seal member SW5 is largely elastically deformed, even when the clearance value of the gap GP5 is changed, the third seal member SW5 is deformed following the change. Although not illustrated, the third seal member SW5 is formed along the upper end 61g of the second combiner 103b and terminates at an end portion of the upper end 61g. The third seal member SW5 reliably prevents the moisture WA1 entering between the combiner 103 and the shade 104 from passing through the gap GP5, and prevents the moisture WA1 from going around onto the barrel 41 that is the optical case CA.

(97) In this modified example, the image display device 200 includes a fourth seal member SW6 between the barrel 41 which is the optical case CA and the lower case CV2. Specifically, the gap GP4a is sealed by arranging the fourth seal member SW6 formed of an elastic body to be sandwiched between a protrusion formed at a rear lower portion of the barrel 41 and a lower end portion of the rear protruding portion 73t of the lower cover 73. The fourth seal member SW6 is fixed to one of the barrel 41 and the lower cover 73 with an adhesive. In the fourth seal member SW6, a surface that is not fixed is smooth, and allows displacement or movement of the barrel 41 or lower cover 73 that faces the surface. Since the fourth seal member SW6 is largely elastically deformed, even when the clearance value of the gap GP4a is changed, the fourth seal member SW6 is deformed following the change. Although not illustrated, the fourth seal member SW6 is formed along the rear protruding portion 73t of the lower cover 73 and extends to the left and right end portions of the rear protruding portion 73t.

(98) The image display device 100 of the modified example illustrated in FIG. 22 may be a fixed type in which the optical device 101 is not rotated. When the optical device 101 is not rotated, the

bearing member **59a** illustrated in FIG. 5 is not required, and the gaps **GP4** and **GP5** can be narrowed or the gaps **GP4** and **GP5** can be eliminated. Since the clearance value does not change even when the gaps **GP4** and **GP5** remain, the elastic deformation of the third seal member **SW5** and the fourth seal member **SW6** is not essential.

(99) The image display device **100** according to the embodiment described above includes the display elements **11a** and **11b**, the optical members **2a** and **2b** for imaging, the circuit members **80a** and **80b** serving as the circuit board **MZ**, the optical case **CA** accommodating the display elements **11a** and **11b** and the optical members **2a** and **2b**, the metallic frames **52a** and **52b** supporting the circuit members **80a** and **80b** serving as the circuit board **MZ** from below, and the exterior case **7** accommodating the optical case **CA** and the metallic frames **52a** and **52b**, wherein the metallic frames **52a** and **52b** are disposed above the optical case **CA**, and the exterior case **7** includes the upper case **CV1** and the lower case **CV2** and includes the first seal member **SW1** at the first facing portion **FC1** between the upper case **CV1** and the lower case **CV2**.

(100) In the image display device **100**, since the first seal member **SW1** is provided in the first facing portion **FC1** between the upper case **CV1** and the lower case **CV2**, the gap between the upper case **CV1** and the lower case **CV2** can be closed in the exterior case **7**. In the exterior case **7**, since the circuit board **MZ** is disposed above the optical case **CA**, the circuit board **MZ** is not disposed in the vicinity of a water intrusion path. Even when water enters from between the exterior case **7** and the optical case **CA**, it is possible to prevent water from entering the circuit board **MZ** disposed on the inner side of the exterior case **7**.

Modified Examples and Others

(101) Although the present disclosure has been described with reference to the above embodiments, the present disclosure is not limited to the above embodiments and can be implemented in various modes without departing from the spirit of the disclosure. For example, the following modifications are possible.

(102) Although the HMD **200** includes the first display device **100A** and the second display device **100B** in the above description, the HMD **200** or the image display device **100** may be configured so that a single first display device **100A** or second display device **100B** is supported in front of the eyes by the support device **100C**.

(103) The first metallic frame **52a** and the second metallic frame **52b** are not limited to a magnesium alloy, and can be formed of an aluminum alloy or other metals. Specifically, the first metallic frame **52a** and the second metallic frame **52b** may be formed of an alloy including one or more of manganese, aluminum, and titanium. The joint **50c** may also be formed of an alloy including one or more of manganese, aluminum, and titanium.

(104) The lower case **CV2** does not need to be divided into the middle frame **72** and the lower cover **73**, and they can be formed as one component.

(105) The shape and length of the gap **GP3** which is the first gap and the gap **GP4** which is the second gap can be appropriately changed according to the use of the HMD **200** or the like.

(106) Components such as the camera **3a**, the illuminance sensor **3b**, the proximity sensor **3c**, and the speaker can be omitted as appropriate according to the application. A waterproof structure may be provided around the speaker.

(107) Although the case in which the first display unit **20a** is screwed to the first metallic frame **52a** has been described above, the first display unit **20a** or the barrel **41** can be fixed to the first metallic frame **52a** not only by screwing but also by caulking, bonding, or the like. In this case, the number of fixing locations is not limited to four, and various fixing locations can be provided, and linear or planar fixing extending continuously is also possible.

(108) The first projection optical system **12a** and the first combiner **103a** can be coupled and fixed to the first metallic frame **52a** to be suspended therefrom. In this case, it is not necessary to fix the first combiner **103a** to the first projection optical system **12a** or the barrel **41**. As a method of suspending the first combiner **103a** from the first metallic frame **52a**, a method similar to the

method used when the first projection optical system **12a** or the barrel **41** is suspended from the first metallic frame **52a** can be used.

(109) In the description above, although it was assumed that the image display device **100** was worn on the head and was used, the image display device **100** may also be used as a hand-held display that is not worn on the head and is to be looked into like binoculars. In other words, the head-mounted display also includes a hand-held display in the present disclosure.

(110) An image display device according to a specific aspect includes a display element, an optical member for imaging, a circuit board, an optical case accommodating the display element and the optical member, a metallic frame supporting the circuit board from below, and an exterior case accommodating the optical case and the metallic frame, wherein the metallic frame is disposed on the upper side of the optical case, and the outer case has an upper case and a lower case, and has a first seal member at a first facing portion between the upper case and the lower case.

(111) In the image display device, since the first seal member is provided in the first facing portion between the upper case and the lower case, the gap between the upper case and the lower case can be closed in the exterior case. The circuit board is not disposed in the vicinity of the water intrusion path by arranging the circuit board above the optical case in the exterior case. Even when water enters between the exterior case and the optical case, it is possible to prevent water from entering the circuit board disposed on the inner side of the exterior case.

(112) In an image display device according to a specific aspect, a lower case includes a middle frame and a lower cover, the middle frame is disposed between an upper case and a lower cover, an exterior case includes a first seal member at a first facing portion between the upper case and the middle frame, and the exterior case includes a second seal member at a second facing portion between the lower cover and the middle frame. Since the lower case is divided into two members and providing the first and second seal members in the first and second facing portions, it is possible to correspond to the shape of the optical case or the like while the sealing property of the exterior case is maintained.

(113) In an image display device according to a specific aspect, the optical case is sealed. In this case, the waterproofing of the image display device can be improved.

(114) In an image display device according to a specific aspect, a metallic frame is supported by a bearing member to be rotatable with respect to an exterior case, and a lower case includes a support portion that supports the bearing member. In this case, visibility can be improved by allowing the rotation of the optical case with respect to the exterior case.

(115) An image display device according to a specific aspect includes a combiner supported by a metallic frame via an optical case, a first gap that is partially bent or narrowed is provided between the combiner and the optical case, and a second gap that is partially bent or narrowed is provided between the optical case and the lower case. It is possible to sufficiently allow the rotation of the combiner by having the first and second gaps. Even when a gap is provided to improve visibility, the first and second gaps that are partially bent or narrowed make it difficult for water to enter. Thus, it is possible to achieve both a waterproof structure and an improvement in visibility in the image display device.

(116) In an image display device according to a specific aspect, the first gap has a curved portion narrowed at the upper end, and the second gap has a curved portion along the exterior of the optical case on the pupil position side of the lower case.

(117) An image display device according to a specific aspect includes a combiner supported by a metallic frame via an optical case, and a shade disposed outside the combiner and supported by an upper case, and further includes a third seal member between the combiner and any of the shade, the upper case, and the lower case. In this case, the waterproof property between the combiner and the lower case can be further enhanced.

(118) An image display device according to a specific aspect includes a shield member that covers a circuit board from above. In this case, the circuit board is sandwiched between the shield member

and the metallic frame, and the waterproof property of the circuit board can be enhanced while electromagnetic interference is prevented by the shield member.

(119) In an image display device according to a specific aspect, the lower cover is formed of a metal.

(120) An image display device according to a specific aspect includes a first optical case for a right eye corresponding to an optical case, a second optical case for a left eye having the same structure as the first optical case, a first metallic frame corresponding to the metallic frame, a second metallic frame supporting the second optical case, and a joint coupling the first metallic frame and the second metallic frame, wherein the joint is covered at an upper side thereof with a shield member and at a lower side thereof with a peripheral seal member. In this case, the joint and periphery thereof can be waterproofed.

(121) In an image display device according to a specific aspect includes a first optical case for the right eye corresponding to an optical case and a second optical case for the left eye having the same structure as the first optical case, wherein the metallic frame is an integrated member that supports the first optical case and the second optical case. In this case, the shape of the metallic frame can be simplified, and waterproofing of the metallic frame and the periphery thereof is simplified.

Claims

1. An image display device comprising: a display element; an optical member for imaging; a circuit board; an optical case configured to accommodate the display element and the optical member; a metallic frame configured to support the circuit board from below; and an exterior case configured to accommodate the optical case and the metallic frame, wherein the metallic frame is disposed above the optical case, the exterior case includes an upper case and a lower case, and includes a first seal member at a first facing portion between the upper case and the lower case, the lower case includes a middle frame and a lower cover, the middle frame is disposed between the upper case and the lower cover, and the circuit board is disposed between the middle frame and the lower cover, wherein the exterior case includes the first seal member at the first facing portion between the upper case and the middle frame, and the exterior case includes a second seal member at a second facing portion between the lower cover and the middle frame.

2. The image display device according to claim 1, wherein the optical case is sealed.

3. The image display device according to claim 2, wherein the metallic frame is supported by a bearing member to be rotatable with respect to the exterior case, and the lower case includes a support portion that supports the bearing member.

4. The image display device according to claim 1, further comprising: a combiner supported by the metallic frame via the optical case; a first gap that is partially bent or narrowed between the combiner and the lower case; and a second gap that is partially bent or narrowed between the optical case and the lower case.

5. The image display device according to claim 4, wherein the first gap has a curved portion that is narrowed at an upper end thereof, and the second gap has a curved portion along an exterior of the optical case on a pupil position side of the lower case.

6. The image display device according to claim 1, further comprising a combiner supported by the metallic frame via the optical case; a shade disposed outside the combiner and supported by the upper case; and a third seal member between the combiner and any of the shade, the upper case, and the lower case.

7. The image display device according to claim 1, further comprising a shield member configured to cover the circuit board from above.

8. The image display device according to claim 1, wherein the lower cover is formed of a metal.

9. The image display device according to claim 1, further comprising: a first optical case for a right eye corresponding to the optical case; a second optical case for a left eye having the same structure

as the first optical case; a first metallic frame corresponding to the metallic frame; a second metallic frame configured to support the second optical case, and a joint configured to couple the first metallic frame and the second metallic frame, wherein an upper side of the joint is covered with a shield member and a lower side thereof is covered with a peripheral seal member.

10. The image display device according to claim 1, further comprising: a first optical case for a right eye corresponding to the optical case; and a second optical case for a left eye having the same structure as the first optical case, wherein the metallic frame is an integrated member that supports the first optical case and the second optical case.
